Enabling Onboard Navigation for UAVs in GPS Denied Environments

MIP400A : B.Tech Project - MidTerm Evaluation

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Objectives

Goal: give a UAV a way to know its own position when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors and physical measurements of its surroundings.

- **This semester:** get an end-to-end navigation pipeline *working* in simulation (image/data in, position out).
- **Next two semesters (through July 2027):** make the pipeline *fast and power-efficient* enough to run onboard real flight hardware.

Core approach: Scene-Aided Navigation

Replace the satellite signal with passive measurements the aircraft can already sense, such as terrain, gravity, magnetic field, and visual scene, matched against a reference map. This turns navigation into a *physical measurement + estimation* problem, much like classical inertial guidance.

Motivation

- GNSS is a **weak radio signal** broadcast from space: it is easily jammed, spoofed, or simply unavailable (urban canyons, mountains, contested airspace, indoors).
- Autonomous UAVs used in surveying, disaster response, and defense all need to keep flying *when GPS fails*, not just when it works.
- **Mechanical perspective:** a UAV is fundamentally a mechanical system in motion. Its onboard IMU already measures physical quantities such as acceleration, rotation, and gravity using mechanical/MEMS sensors. The question this project asks: can these existing physical measurements, combined with a camera, replace the satellite link entirely?
- **Precedent:** TERCOM and DSMAC, terrain and scene matching systems developed for Cold-War cruise-missile guidance, proved decades ago that passive, GPS-free navigation using physical terrain data is reliable and flight-proven.

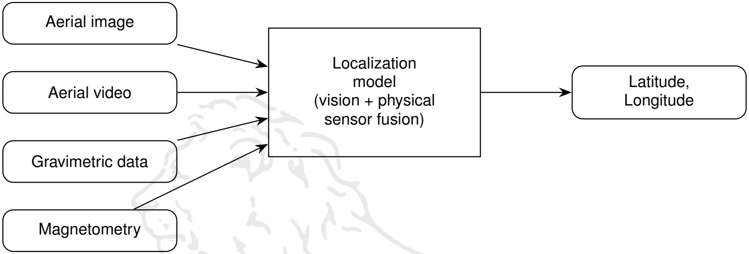
Background of Similar Work

Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

Method	Physical quantity measured	Passive?	Assessment
Terrain matching (TERCOM)	Ground elevation	Yes	Proven, strong candidate
Visual landmark matching	Camera imagery	Yes	Strong if paired with AI matching
Gravity matching	Gravity anomalies	Yes	Promising, needs sensitive sensor
Gravity-gradient nav.	$\partial g / \partial x, y, z$	Yes	Very promising, higher precision
Magnetic anomaly nav.	Magnetic field	Yes	Useful in mountainous terrain
Quantum inertial sensing	Acceleration / rotation	Yes	No map needed; pure mechanics
Wi-Fi / cellular / RF	Radio signal strength	No	Rejected: unavailable at altitude

Existing implementations reviewed: VGG16 and CLIP-based deep feature matching of drone frames to satellite tiles (sub-3–7 m RMSE), classical SIFT/ORB feature matching, terrain-weighted optimization against DEM maps.

Proposed Work



- **Current phase (no drone hardware yet):** feed pre-recorded aerial imagery into the pipeline to validate feasibility before committing to flight tests.
- **Immediate next step:** replicate a proven reference implementation (satellite-image matching network) in simulation as our working baseline.
- **Following step:** fuse in gravimetric and magnetic measurements for redundancy, then optimize the whole pipeline to run in real time on embedded flight hardware.

Timeline of the Work

Phase 1: Now to Dec 2026 (this semester)

Replicate a reference vision-matching implementation; get an end-to-end simulated pipeline (aerial image $\rightarrow$ position estimate) working.

Phase 2: Jan 2027 to Apr 2027

Fuse in gravimetric and magnetic sensing for redundancy; stress-test accuracy across varied terrain and lighting conditions.

Phase 3: May 2027 to July 2027

Optimize the pipeline for speed and power efficiency; integrate onto embedded flight hardware for a physical UAV test.